TAEKJUN (TJ) LEE

Business Systems & Automation Engineer

Seoul, Republic of Korea | Open to relocation: New Zealand & Australia | Employer sponsorship required

leetj94@gmail.com | LinkedIn | Portfolio & case studies

PROFILE

Business systems engineer at Samsung Electronics, combining firsthand facilities engineering experience with hands-on software delivery. Translates operational needs into data standards, integrated workflows and Python/Vue applications. Develops across AWS and on-premises environments, with responsibility for requirements, implementation, deployment and production support.

PROFESSIONAL EXPERIENCE

Samsung Electronics | Device Solutions | Jan 2019 - Present

Systems Engineer | 2020 - Present

- Analyse engineering workflows and develop business systems, data integrations and automation for semiconductor infrastructure operations.
- Develop Python/FastAPI backends, Vue.js interfaces and ETL/EAI integrations; deliver and support applications on AWS and on-premises infrastructure.
- Work with engineering users and vendors on requirements, technical evaluation and adoption. Supported Austin and Taylor FAB rollouts during a U.S. assignment, Sep-Nov 2024.

Facilities Engineer | 2019 - 2020

- Maintained and operated infrastructure supply equipment. Analysed equipment quality and maintenance data as an FDC/PM conductor to investigate anomalies and component replacement intervals.

SELECTED SOFTWARE & AUTOMATION WORK

Engineering Document Delivery Platform | 2025 - 2026

Project owner, process analysis, full-stack development and AWS delivery

- Replaced email-based contractor handoffs with direct submission, automated extraction and rule validation, correction feedback and final employee approval.
- Built FastAPI/Vue web and backend services. Connected S3, EventBridge, SQS, Lambda and RDS; integrated Autodesk APS Model Derivative and Design Automation APIs.
- Implemented retries, duplicate detection and idempotent ingestion. Configured EC2 delivery and CodeCommit/CodeDeploy/CodePipeline CI/CD with operational checks and rollback procedures.
- Project scope: approximately 30,000-70,000 drawings and 10,000 registered assets/components per FAB.

3D Digital Twin Platform | 2023 - 2024; ongoing extensions

Project owner, vendor evaluation, Vue/Python development and on-premises deployment

- Built application and integration layers connecting eight operational systems in a searchable 3D workspace, linking approximately 1.5 million assets/components for approximately 1,000 users.
- Connected equipment location to quality trends, maintenance history and work status. Evaluated NVIDIA, Siemens and AVEVA technologies through meetings and PoCs.
- Developed Vue interfaces, Python backend and ETL/EAI; delivered with Git, Jenkins and Kubernetes-based CI/CD. Current extensions include simulation model conversion and AMR/AGV spatial-data integration.

SELECTED WORK CONTINUED

Facility Master Data Platform | 2020 - 2021

Domain specialist, data/process analyst, validation and integration developer

- Defined common equipment data standards and lifecycle workflows across quality, maintenance, operations and work systems, accounting for different registration timing and business requirements.
- Prepared approximately one million rows across 50 columns. Developed Python/SQL pre-registration checks, ETL/EAI integrations and Spotfire reconciliation reports using SQL across system databases.
- Reduced engineer time for equivalent cross-system registration work by 87%; measured cross-system discrepancy rates fell from 17% to 0%.
- Supported production requests, incidents and training for approximately one year; extended features and integrations, including environmental safety and material-registration systems.

Drawing Management & Search | 2021 - 2022

- Redesigned drawing handover processes, standardised drawing objects and conventions, and structured equipment-to-drawing relationships so engineers could search by equipment instead of opening files individually.
- Designed extraction and integration workflows and automated completeness/design-rule checks for a platform serving approximately 4,500 users.

Data Platforms, Governance & Operational Automation | 2023 - 2026

- Managed data from eight infrastructure systems for a data lake initiative. Defined metadata, security classifications and disclosure/access rules.
- Built a FastAPI/Oracle/Vue.js ETL operations dashboard with job-level logs, failure alerts and retry controls. Taught SQL and Spotfire within an engineering organisation of approximately 2,000 employees.
- Contributed to AI-ready data eligibility work in 2026. Authored reusable team guidance for API standards, UI design, testability and fallback behaviour; prototyped workflow/data integrations using external language models.

TECHNICAL SKILLS

Development: Python, SQL, JavaScript, FastAPI, Vue.js, REST APIs, Pydantic, pytest

Data & integration: Oracle, PostgreSQL, MySQL/MariaDB, SQL Server, ETL/EAI, data modelling and quality, Spotfire

Delivery & operations: AWS EC2/S3/Lambda/SQS/EventBridge/RDS, Linux, Nginx, Git, Jenkins, Kubernetes, CI/CD

Engineering systems: MES/MIS, Autodesk APS/ACC, Revit/BIM, equipment data standards, process analysis

EDUCATION & LANGUAGES

M.S., Industrial Engineering | Sungkyunkwan University | Feb 2026 | GPA 4.38/4.5

B.S., Mechanical Engineering | Sungkyunkwan University | Feb 2019

Korean: native | English: TOEFL iBT 107/120 (Apr 2025)

RECOGNITION

Samsung Best Contributor/Performer (2022, 2023). Project Incentive Awards: Master Data (2021), Drawings (2022), Digital Twin (2023), 3D Integration (2024).